Large gauge fields: action cost, constrained minima and model entropy

Maintained conclusions. For a smooth finite-action continuum connection whose action-decreasing flow exists through $t = \ell^2/8$ with no boundary flux, a flowed block constraint $\int_B |G_t|^2 \geq \eta_F^2$ implies $S_E/\hbar \geq \eta_F^2/(4g^2)$ in the declared ordered-index norm. Equality has explicit necessary and sufficient conditions; no admissible positive-threshold saturating family is constructed.

The constant-background unstable-mode count is a separate phase-space model. Its charge and real/complex multiplicities give explicit ratios, not a normalized interacting entropy bound. Finite-regulator compactness gives a constrained minimum without a coercive Gaussian Hessian. None of these statements proves local rarity, polymer convergence or a physical spectral gap. The former claimed convergence threshold is replaced by its actual unproved weight assumption below. Mathematical source/model checks are internal.

Continuum action cost and equality

Take $\Omega = \mathbb{R}^4$ with sufficient decay for the identities below, or a finite periodic four-dimensional domain, and $B \subset \Omega$ a cube of side ℓ. Other boundary conditions are allowed only if they justify the same dissipation identity without a boundary flux. All connections under consideration have smooth flow B_s on $0 \leq s \leq t$ and finite action; its curvature is G_s, with $G_0 = F$.

At $\sqrt{8t} = \ell$, define

$$\mathcal{F}_\ell(A) = \ell^{-4} \int_B |G_t|^2 d^4x, \qquad K_\eta = \{A : \mathcal{F}_\ell(A) \geq \eta^2/\ell^4\}.$$

The constraint is gauge invariant. It measures a flowed block L^2 norm. It does **not** give a supremum or an adjoint-curvature operator norm bound. In particular, it cannot be substituted for the curvature upper bound used in the Jacobian estimates of the Jacobian note or the corrected instability note, §1 (relevant passages read). The original phrase "up to replacing a supremum by a block average" concealed an additional estimate; that identification is withdrawn.

The bound and its equality conditions

Proposition 1 (continuum action cost). Under the assumptions of Section 1, for every $\ell > 0$ and $\eta \geq 0$,

$$s(A) \geq \frac{\eta^2}{4g^2} \qquad (A \in K_\eta).$$

The infimum has the same lower bound; if the set is empty its infimum is understood as $+\infty$.

Proof. Use the ungauged flow

$$\partial_s (B_s)_\nu = D_\mu (G_s)_{\mu\nu} =: J_{s,\nu}.$$

Variation of $s = (4g^2)^{-1} \int \sum_{a,\mu,\nu} (G^a_{\mu\nu})^2$ and integration by parts give

$$\delta s = -\frac{1}{g^2} \int \sum_\nu \langle J_{s,\nu}, \delta B_\nu \rangle d^4x, \qquad \frac{d}{ds} s(B_s) = -\frac{1}{g^2} \|J_s\|_2^2 \leq 0.$$

The norm in the last display sums the colour and one-form indices. The factor $1/g^2$ is explicit for this flow-time convention; the old formula $dS_E/ds = -\|D^* G\|_2^2$ omitted the action prefactor. The continuum flow convention is the one recorded in Lüscher's local companion (introductory flow equations, passage read); the displayed variation checks the normalization directly.

Consequently

$$s(A) \geq s(B_t) = \frac{1}{4g^2} \int_\Omega |G_t|^2 \geq \frac{1}{4g^2} \int_B |G_t|^2 \geq \frac{\eta^2}{4g^2}.$$

The second inequality is positivity outside B; the last is the constraint. □

Scope of uniformity. The same algebraic bound applies at each admissible ℓ and domain volume; it does not assert flow existence or regularity uniformly through any limit. It uses a fixed invariant inner product and action coefficient. Writing the action in this form for different compact gauge groups gives the same coefficient by **normalization convention**, not a comparison theorem for their measures, mode counts or operators. Rescaling the inner product also rescales the field norm and the threshold. The former claim for every lattice spacing $a < \ell$ is withdrawn: an exact Wilson-lattice analogue requires a specified lattice observable, its action comparison and its flow monotonicity, none of which is proved by this continuum integral.

Proposition 2 (equality accounting; replacement for withdrawn sharpness). For every $A \in K_\eta$ in Proposition 1,

$$s(A) - \frac{\eta^2}{4g^2} = \frac{1}{g^2}\int_0^t \|J_s\|_2^2 ds + \frac{1}{4g^2}\int_{\Omega\setminus B} |G_t|^2 d^4x + \frac{1}{4g^2}\left(\int_B |G_t|^2 d^4x - \eta^2\right).$$

Each term is nonnegative. Equality therefore holds if and only if all three vanish: zero flow dissipation, zero flowed curvature outside B, and exact saturation of the block threshold.

Proof. Integrate the dissipation identity, split the final action integral into B and its complement, and subtract the threshold. Nonnegativity gives the equivalence. For a smooth flow, zero dissipation entails $J_s = 0$ throughout the interval. □

This is an identity and necessary-and-sufficient equality test, **not** an existence assertion. For $\eta = 0$ a flat connection gives equality when admitted by the domain. For $\eta > 0$, neither an equality example nor a sequence approaching equality is constructed here, so the optimality of $1/4$ remains unproved. Strict inequality for particular fields would not itself give a uniform improvement of the infimum.

Why the old example fails. Prescribing a nonzero constant abelian curvature on B and zero outside does not produce a smooth admissible connection. In that sector the Bianchi identity is $dF = 0$, whereas $d(\mathbf{1}_B F_{\text{const}})$ has nonzero face-supported terms for a nonzero constant two-form on a bounded cube. A smooth cutoff of the connection can give an admissible localized field, but then its curvature includes transition terms whose cost and flow must be controlled. No such control is supplied by the former example.

Nor does a/ℓ control flow at the radius of the block. Already in the abelian heat-flow model, rescaling $x = \ell y$ leaves the smoothing time $t/\ell^2 = 1/8$ fixed. The heat kernel samples a boundary layer of width comparable with ℓ, so a relative effect of order one is possible as $a/\ell \to 0$. There is no vanishing cutoff correction justified by that limit. These observations invalidate the old sharpness proof without claiming that no other sharpness proof exists.

Constrained minima and their tangent conditions

Proposition C (finite regulator only). Let G be compact, let $\mathcal{E}$ be a finite link set, and let $K \subset G^{\mathcal{E}}$ be closed and nonempty. Every continuous action s attains its minimum on K. If s is real valued, its integral $\int_K e^{-s} d\nu$ is finite for every finite reference measure ν.

Proof. The product and its closed subset are compact. The extreme value theorem gives a minimum and a finite maximum; e^{-s} is bounded, so its integral against a finite measure is finite. □

This says nothing about regulator-uniform estimates, the size of a Gaussian contribution or convergence of a saddle-point expansion. For a C^2 action, a constrained minimum requires nonnegative first variation along one-sided feasible curves. At a boundary minimum the first variation need not vanish, so the Hessian need not be nonnegative on the tangent cone. In a local coordinate, take

$$K = [0, 1/4], \qquad s(x) = x - x^2.$$

The minimum is at 0, with $s'(0) = 1$ but $s''(0) = -2$, even though positive directions belong to the tangent cone. Along a twice differentiable feasible curve γ whose first action derivative vanishes, the necessary condition is

$$(s \circ \gamma)''(0) = \operatorname{Hess} s[\dot\gamma, \dot\gamma] + ds[\ddot\gamma] \geq 0$$

in local coordinates. Constraint curvature can matter as well. At an interior minimum $ds = 0$ and the Hessian is nonnegative, but it can have zero modes: $s(x) = x^4$ has no coercive quadratic term at its minimum. Gauge degeneracy, boundary geometry, higher terms and uniform determinant bounds need separate treatment.

The corrected instability note, §2 (passage read) supplies a negative quadratic mode for a stationary constant background in its specified background-transverse sector. It excludes that background as a constrained minimum only if an admissible perturbation realizes the descent while remaining in K. No such feasibility proof, or construction of an inhomogeneous minimizer, is supplied here.

A constant-background mode count

Use the continuum background-covariant operator of the instability note, §2 (passage read), in four Euclidean dimensions. For one charged complex field, with charge magnitude $q > 0$ relative to T^3, its transverse polarizations have eigenvalues

$$\omega^2 = k_3^2 + k_4^2 + (2n+1)qb \mp 2qb.$$

Only $n = 0$ with aligned spin is negative, for $k_3^2 + k_4^2 < qb$. Neutral components have no such branch.

Landau degeneracy. The transverse density per unit area is $qb/(2\pi)$. For example, in Landau gauge with transverse length ℓ and periodic second coordinate, momenta are spaced by $2\pi/\ell$. Guiding centres are therefore spaced by $2\pi/(qb\ell)$, giving that density. On a magnetic torus with compatible flux, $qb\ell^2/(2\pi)$ is an integer degeneracy. For a block cut from the plane it is a bulk density, not an exact boundary-condition-independent count.

Longitudinal count. Replacing a momentum sum by its phase-space integral gives

$$\left(\frac{\ell}{2\pi}\right)^2 \operatorname{area}\{k_3^2 + k_4^2 < qb\} = \frac{qb\ell^2}{4\pi}.$$

Thus the **phase-space estimate**, in complex dimension, is

$$\mathcal{N}_{\mathbb{C}}^{\mathrm{ps}}(q) = \frac{q^2 b^2 \ell^4}{8\pi^2} = \frac{q^2 \eta_b^2}{8\pi^2}.$$

An exact periodic longitudinal count, when the magnetic flux and boundary conditions admit this background, would instead be

$$\mathcal{N}_{\mathbb{C}}^{\mathrm{box}}(q) = m_q \, \#\{n \in \mathbb{Z}^2 : 4\pi^2 |n|^2 < q\eta_b\}, \qquad m_q = \frac{q\eta_b}{2\pi} \in \mathbb{N}.$$

The disc area is an approximation to this integer count, justified as $q\eta_b \to \infty$: unit squares around the integer points place it between disc areas with radii differing by a bounded constant. It is not justified merely by sending $\ell/a \to \infty$ at fixed η_b. A lattice operator also has dispersion and cutoff effects; no uniform lattice replacement is proved here.

Real and colour multiplicities. A charged pair of real adjoint components forms one complex field. Each complex mode has two real amplitudes, so its real negative-space dimension is twice the complex count. Take the usual $SU(3)$ generator

$$T^3 = \operatorname{diag}(1/2, -1/2, 0).$$

The matrix units satisfy $[T^3, E_{ij}] = (t_i - t_j)E_{ij}$, where t_i are its diagonal entries. The three conjugate root pairs therefore have charge magnitudes $1, 1/2, 1/2$. Each pair is counted once as a complex field; adding its conjugate again would double-count it. Consequently

$$\mathcal{N}^{\mathrm{ps}}_{\mathbb{C},SU(3)} = (1 + 1/4 + 1/4)\frac{\eta_b^2}{8\pi^2} = \frac{3\eta_b^2}{16\pi^2},$$
$$\mathcal{N}^{\mathrm{ps}}_{\mathbb{R},SU(3)} = \frac{3\eta_b^2}{8\pi^2}.$$

These are estimates for this background and operator sector, not a mode count for every large-field configuration or a gauge-fixed functional determinant calculation.

Action versus model count: the norm and multiplicity matter

The continuum action inequality above uses the ordered-index L^2 threshold η_F; the spectral model instead uses $\eta_b = b\ell^2$.

The spectral parameter $\eta_b = b\ell^2$ of Section 2 is not implicitly η_F. To compare them in the homogeneous model, fix the colour norm by $\langle T^3, T^3\rangle = 1$ (equivalently $2\,\mathrm{tr}[(T^3)^2] = 1$ in the Hermitian convention), and absorb the coupling into the curvature. With $F_{12} = bT^3$, $F_{21} = -bT^3$ and all other components zero,

$$|F|^2 = 2b^2, \qquad \eta_F^2 = 2\eta_b^2, \qquad s_B := \frac{1}{4g^2}\int_B |F|^2 = \frac{\eta_b^2}{2g^2}.$$

Here s_B is the block action of the background, not the minimum action for an arbitrary constrained field. No finite-action localization or admissible boundary construction follows from this calculation.

Proposition M (model ratios, not entropy bounds). For $\eta_b > 0$ and the preceding normalization, division by the phase-space counts of Section 2 gives the following identities:

count convention	phase-space count	$s_B/\mathcal{N}^{\mathrm{ps}}$
one unit-charge complex field	$\eta_b^2/(8\pi^2)$	$4\pi^2/g^2$
its real directions	$\eta_b^2/(4\pi^2)$	$2\pi^2/g^2$
all three $SU(3)$ charged pairs, complex	$3\eta_b^2/(16\pi^2)$	$8\pi^2/(3g^2)$
all three $SU(3)$ charged pairs, real	$3\eta_b^2/(8\pi^2)$	$4\pi^2/(3g^2)$

Proof. Substitute the counts and $s_B = \eta_b^2/(2g^2)$, and cancel η_b^2. □

The original arithmetic

$$\frac{\eta_b^2/(4g^2)}{\eta_b^2/(8\pi^2)} = \frac{2\pi^2}{g^2}$$

remains a formal identity if the numerator is an **assumed** cost s_{old}. It paired the ordered-index L^2 threshold formula with the different spectral threshold without their conversion. Under the normalization above, the same number occurs for one pair's real directions, not its complex count. None of these ratios is a universal entropy coefficient. Scale cancellation at fixed η_b and g does not establish cutoff uniformity or running-coupling control.

The missing weight estimate. For a finite regulator with reference probability measure ν and $d\mathbb{P} = Z^{-1}e^{-s}d\nu$, an action floor $s \geq Q$ on an event K gives only

$$\mathbb{P}(K) \leq Z^{-1}e^{-Q}\nu(K).$$

The denominator needs control relative to the constrained numerator. The negative eigenspace dimension at one saddle controls neither this ratio nor the volume of a region, its nonlinear

integration, or the number of possible regions. Compactness at a fixed regulator provides no uniform factor per mode.

To preserve the former calculation's scope explicitly, **if one assumed** a weight bound

$$W(\eta_b) \le e^{-s_{\mathrm{old}}} e^{C\mathcal{N}_{\mathbb{C}}^{\mathrm{ps}}(1)}, \qquad s_{\mathrm{old}} = \frac{\eta_b^2}{4g^2}, \qquad C > 0,$$

then algebra would give

$$W(\eta_b) \le \exp\left[-\frac{\eta_b^2}{4g^2}\left(1 - \frac{Cg^2}{2\pi^2}\right)\right].$$

Its exponent is negative for $g^2 < 2\pi^2/C$. Neither the assumed weight estimate nor a regulator-independent C is proved. The old numerical threshold $g^2 < 2\pi^2$ additionally set $C = 1$ without justification. This display is **conditional bookkeeping**, not a convergence theorem. Even an actual tail in strength would need region-counting and interaction estimates to sum polymers.

Consequence for STATE

Retain the continuum inequality and equality test, the Bianchi counterexample to the former saturation construction, the finite-regulator tangent-cone counterexample and the explicitly normalized charged-background counts. A block L^2 threshold supplies no upper curvature norm. The flow's action descent also compares times on one field, not block functionals of two initial fields. Normalized local weights, boundary-uniform integration and polymer summation remain open. The operator comparison needs separate sign and localization estimates even if a measure bound is subsequently established.